

Deepak Mallampati

FORWARD-DEPLOYED ENGINEER (FDE)

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Kent, OH

PROFESSIONAL SUMMARY

Customer-facing AI engineer with 4+ years end-to-end ownership across data, agents, and production reliability. Shipped customer-facing AI to 20+ internal teams at Vanguard (100K+ req/day, 99.9% uptime); built fault-tolerant data pipelines (-60% failed runs); production stack across SQL/Postgres, Spark/Kafka/Databricks, Python + TypeScript. Comfortable translating non-technical user needs into deployed workflows.

SKILLS

Customer-Facing Delivery • SQL + PostgreSQL • Data Pipelines (Spark/Kafka) • TypeScript + React • Agent Orchestration • Python (4+ yrs) • Production Reliability • End-to-End Ownership • Multi-Tenant Integrations • Workflow Translation

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), Java, C++

AI / ML: PyTorch, scikit-learn, XGBoost, LangChain, LlamaIndex, OpenAI, Anthropic, Hugging Face Transformers, Diffusers, YOLOv8

LLM / GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search

Backend: FastAPI, Flask, REST, WebSockets, Docker, Celery, Playwright

Frontend: React, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)

Data: PostgreSQL + pgvector, MongoDB, Redis, Pandas, NumPy, Snowflake

DevOps: Docker, Jenkins, GitHub Actions, Grafana, Prometheus, AWS (EC2, S3, Lambda)

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Agent Engineering, Production LLM

USA - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval; recursive semantic chunking and transformer embeddings improved retrieval precision **~35%**, reduced irrelevant retrieval **>30%**, response alignment **>90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries with tool discipline, structured reasoning, and grounding rules; response stability **~25%**, hallucinations cut **>30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show and care engagement scoring; recall **+15-20%** on high-risk categories, precision held **>90%** via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects **~30%**.
- EHR preprocessing (Pandas, NumPy); dataset reliability **>98%**, downstream instability **-40%**.
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas ERP

- Optimized **T-SQL stored procedures** on a compliance-sensitive ERP (contracts, nominations, allocations, invoicing); query time **-20%**, retrieval latency **-25%**.
- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments; deploy errors **-30%**, release cycle **-35 to -40%**.
- Designed performance dashboards with **SQL Server DMVs + Grafana**; incident recurrence **-25%**, root-cause resolution **-30%**.
- Tuned index maintenance and partition-aware queries; reconciliation runtime **-15%**, deadlocks **-15%**.
- Supported RBAC and audit logging for financial modules in a regulated environment.

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization and timestamp-aligned clip extraction cut manual review **60-70%**.
- AI-selected highlights aligned with human-curated moments **~85%**.
- Implemented **document Q&A with RAG**; hallucinations **~30%**, contextual accuracy **>85%**.
- Deployed as **Flask API** with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

5-stage multi-agent pipeline (Intake, Validation, Consistency, Duplicate, Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies.

Audit-ready explainable risk scoring.

Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts

Manga Lens - Multi-Provider AI Vision Chrome Extension

Real-time manga translation extension shipped to the Chrome Web Store. Multi-provider abstraction (4 AI vision providers: Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama) with per-provider payload handling for worst-case failure isolation. 29 sites supported; 7-day cache; privacy policy and narrowed permissions.

TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Ollama

Dream Decoder - Multimodal Creative Platform

End-to-end multimodal pipeline: text description, intermediate prompt-transformation layer, grounded image generation. ~30% contextual alignment gain, ~25-30% first-pass image success.

FastAPI, React + Vite, Tailwind, Perplexity Sonar, Replicate

YOLOv8 Driver Drowsiness Detection

Real-time computer vision for driver drowsiness; sub-50ms inference on edge hardware. Walk-forward validation and class-weighted training for imbalanced safety data.

PyTorch, YOLOv8, OpenCV, ONNX

RESEARCH

Kent State University - ML & Gen AI Research Assistant

Oct 2023 - Jan 2024

Privacy-Preserving ML & LLM Fine-Tuning Research

- Privacy-preserving ML on a clinical hospital-readmission dataset; k-anonymity (k=3), l-diversity (l=2), Laplace differential privacy across 4 ϵ levels (0.1-2.0).
- Reduced record-linkage re-identification risk from **3.38% to 0.32%**; retained **99% of baseline predictive performance** (RF AUC 0.689 vs. 0.696; F1 0.895).
- Composite **privacy-utility scoring framework** benchmarking six anonymization configurations; identified k=3 / l=2 as optimal operating point.
- Fine-tuned **Qwen3-27B via 4-bit QLoRA on NVIDIA H100** with a curated 361-example instruction dataset (six-stage narrative schema).
- Authored peer-review-ready **IEEE-format conference paper** documenting methodology, experiments, and results (LaTeX).

EDUCATION

Master of Science, Computer Science - Kent State University

Aug 2023 - May 2025

Focus: Applied Machine Learning, NLP, Database Systems.

Bachelor of Technology, Computer Science - KL University

Jun 2019 - May 2023

Founder / Lead of E-Farming platform (university project, pilot with 3 cooperatives).

CERTIFICATIONS

Deep Learning Specialization - DeepLearning.AI (Coursera)

2024